Does Explicit Conflict Resolution Help Multi-Agent LLM Trading? A Controlled Comparison with CryptoTrade

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Abstract

LLM trading agents increasingly combine heterogeneous sources such as prices, on-chain statistics and news, which often point in opposite directions. Existing agents such as CryptoTrade fuse all sources inside a single prompt and leave disagreement to be resolved implicitly. We study whether resolving disagreement *explicitly* helps: we assign one specialist agent per source, quantify their disagreement with a KL-divergence and variance score, and, when it is high, run a multi-round debate in which every claim must cite a source line and pass a verification step. To isolate the effect of the architecture, we reimplement CryptoTrade, verify it against the authors' code, and run both systems on the same data, trading environment, information timing and local LLM over the paper's bull, sideways and bear Bitcoin periods of 2023, plus a development set of 145 unused days. The conflict-aware system trails CryptoTrade on both the test periods (+0.38% vs. +5.71% average return) and the development set (+8.60% vs. +11.95%). Our analysis shows that (i) debate lowers the measured conflict from 0.61 to 0.22 yet changes only 11 of 195 decisions, always towards abstention; (ii) each agent's directional accuracy is near chance (45–48%) and biased, with the news agent calling BUY about 80% of the time and the market agent selling into rallies because CryptoTrade's indicator labels are inverted; (iii) on the 170 days the agents conflict, debate resolves the conflict correctly 48% of the time, no better than no debate; and (iv) eight improvements tuned on the development set do not beat CryptoTrade reliably. CryptoTrade itself is at chance at 1–7-day horizons and buys on 91% of its directional days; removing our system's SELLs closes the gap. A pre-registered check on ETH confirms these findings, refutes the one improvement found after seeing the data (an arbiter that trusts agents by their past record lowers accuracy from 52.3% to 46.9%), and reverses the ranking: on ETH test periods our system beats CryptoTrade by 7.9 points. These findings separate *reducing* disagreement from *resolving* it, and show that returns here reflect market exposure more than prediction.[1]

1 Introduction

Cryptocurrency prices respond to signals of different nature: technical momentum, blockchain usage and news. These signals are noisy and frequently disagree, for instance an overbought RSI alongside positive news. Large language models (LLMs) can read numbers and text together, and a growing line of work builds trading agents on top of them (Yu et al., 2023, 2024; Xiao et al., 2024; Li et al., 2024). CryptoTrade (Li et al., 2024), the reference agent for cryptocurrency, asks one LLM to analyse prices and on-chain data, another to analyse news, and a trader LLM to read both analyses and act. When the analyses disagree, the trader resolves the conflict inside one prompt: disagreement is never measured, explained or contested.

Work on multi-agent debate suggests a different design. Letting several LLM instances critique each other improves factuality and reasoning (Du et al., 2024; Liang et al., 2024), and MADAM-RAG (Wang et al., 2025) shows that assigning one agent per retrieved document and letting them debate helps when documents conflict. These results come from question answering with a single correct answer. Whether they transfer to trading, where "correctness" is only revealed by the next price move and the sources are structurally different, is open.

We ask four research questions under strictly controlled conditions (same data, environment, information timing and LLM):

- **RQ1:** Does explicit conflict handling across source-specialised agents outperform CryptoTrade's implicit fusion?

[1] Code and data: https://github.com/Chivu171/Multi-Agent-Crypto

- **RQ2:** What does each source contribute, and does its contribution depend on the market regime?
- **RQ3:** Does debate change decisions, or does it only lower the conflict score?
- **RQ4:** Can improvements motivated by this analysis close the gap when their parameters are chosen on separate development data?

Our contributions are: (1) a conflict-aware multi-agent trading system with a KL–variance conflict score, root-cause analysis and a *grounded* debate in which every claim cites a source line and is verified before it can change the state; (2) a faithful reimplementation of CryptoTrade, verified to match the authors' code exactly on rule strategies and prompts, which makes a controlled comparison possible with a local open model; (3) an empirical study, replicated on a held-out development set, showing that explicit conflict handling trails implicit fusion, that debate reduces disagreement without resolving it, and that individual agents are near chance and directionally biased; (4) a dev-then-test evaluation of eight improvements with all results reported, including the negative ones, and a pre-registered replication on ETH that confirms the main findings and refutes a post-hoc one.

2 Related Work

LLM trading agents. FinMem (Yu et al., 2023) equips a stock-trading agent with layered memory; FinCon (Yu et al., 2024) organises agents as a manager–analyst hierarchy with verbal reinforcement; TradingAgents (Xiao et al., 2024) simulates a trading firm in which bullish and bearish researchers argue before a trader acts. For cryptocurrency, CryptoTrade (Li et al., 2024) combines on-chain statistics and news with a reflection step inspired by Reflexion (Shinn et al., 2023), and Luo et al. (2025) compare hierarchical, collaborative and debate communication among modality-specialised agents for portfolio management. Market Regime Council (Zheng et al., 2026) weights three specialist agents online with Shapley credits and regime-dependent multipliers. Closest to us, CGX (Bajgora et al., 2026) lets Bull and Bear agents debate for three rounds and trades crypto only when a meta-evaluator judges their consensus strong enough, mainly reducing drawdown. None of these measures disagreement *between data sources* or compares against a reproduced baseline under an identical model.

Multi-agent debate and conflicting evidence. Du et al. (2024) show that multiple LLM instances debating over rounds improve reasoning and factuality; Liang et al. (2024) use opposing debaters and a judge to counter degeneration of thought; ChatEval (Chan et al., 2024) applies debate to evaluation. MADAM-RAG (Wang et al., 2025) assigns one agent per document and aggregates the debate history across rounds to handle ambiguity and misinformation in retrieval. Our design follows this one-agent-per-source pattern and adds a quantitative conflict trigger and citation verification. In simulated portfolio allocation, Huang et al. (2026) find that debate reasoning quality is unrelated to Sharpe and identify *sycophantic convergence*, agents abandoning independent positions, as its main failure mode.

3 Method

Figure 1 shows the pipeline. For a decision at the open of day D, three specialists analyse their own source; a Conflict Analyzer measures their disagreement; above a threshold it triggers root-cause analysis (RCA) and debate; a Mediator aggregates the final opinions into a signal.

3.1 Specialist agents

The *Market* agent reads the last seven daily opens, EMA20/50 and RSI14 of opens, and the three technical signals CryptoTrade uses (moving-average crossover, MACD, Bollinger bands). The *Financial* agent reads six on-chain transaction statistics of day $D-1$ with their day-over-day change. The *Sentiment* agent reads the news articles of day $D-1$. Every evidence item (a metric, a news sentence) is placed on its own line so that later stages can cite it precisely. Each agent returns a signal, a belief (d_i, s_i) with direction $d_i \in \{-1, 0, 1\}$ and strength $s_i \in [0, 1]$, a reasoning path, and metadata. A noise score e_i is computed from data rather than by the LLM: $1 - |\text{RSI} - 50|/50$ for Market, the mean absolute daily change of on-chain metrics divided by 30% (capped at 1) for Financial, and a neutral 0.5 for Sentiment since the dataset has no fear-and-greed index.

3.2 Conflict Analyzer

Each belief is mapped to a distribution over {down, up}, $P_i = (0.5 - 0.5x_i,\ 0.5 + 0.5x_i)$ with $x_i = d_i s_i$, so a weak opinion stays close to uniform. The

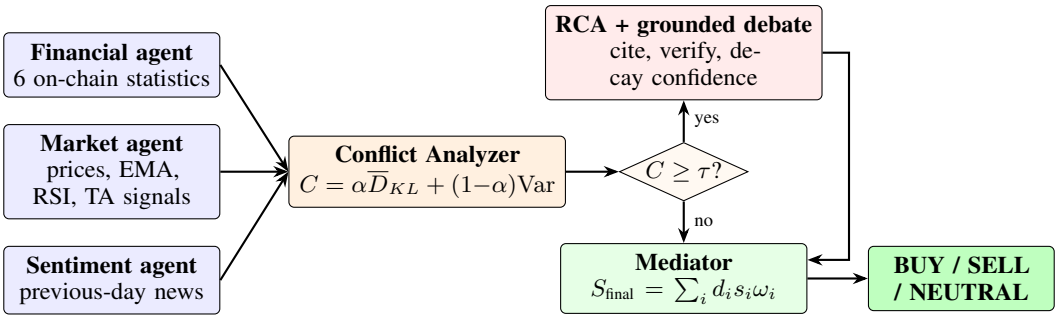


Figure 1: Conflict-aware multi-agent pipeline. Each source has its own agent; disagreement is measured and, above a threshold, explained and debated with verified citations before aggregation.

conflict score combines the mean pairwise KL divergence (Kullback and Leibler, 1951) over ordered pairs with the variance of the signed beliefs:

$$C = \alpha \tfrac{1}{N(N-1)} \textstyle\sum_{i \neq j} D_{KL}(P_i \,\|\, P_j) + (1-\alpha)\,\mathrm{Var}(x_1, \ldots, x_N), \quad (1)$$

with $\alpha = 0.6$ and $N = 3$. When $C \geq \tau = 0.4$, rule-based categories (signal, temporal, reliability, redundancy conflict) are assigned and an LLM writes an RCA of 1–3 cited claims explaining the disagreement.

3.3 Grounded debate

In each of two rounds, agent i receives a rebuttal strength

$$R_i = \mathrm{clip}\Big(\tfrac{1}{1.5(N-1)} \textstyle\sum_{j \neq i} \big[w_b D_{\cos} + w_e D_J + w_l D_L \big],\, 0,\, 1 \Big), \quad (2)$$

averaging the cosine distance between signed beliefs, the Jaccard distance between evidence vocabularies and the normalised Levenshtein distance between reasoning paths ($w_b = w_e = w_l = 1/3$). Its confidence decays as $s_i \leftarrow s_i e^{-\beta R_i}$ with $\beta = 0.35$, and the agent rewrites its reasoning given the accumulated debate history (as in MADAM-RAG's aggregator), all agents' evidence and a constraint to stay within its role. The update changes the *strength* s_i but never the *direction* d_i.

Citation verification. Every RCA and debate output consists of 1–3 claims labelled fact, inference or hypothesis, each citing an evidence id and a line id. Code checks the schema and resolves the quoted line itself, so the model never copies numbers. A separate LLM call then judges each claim as supported, unsupported or contradicted against the full evidence; a failed or missing review counts as rejection. After one retry, a rejected update is rolled back entirely, including the confidence decay, so unsupported arguments cannot move the decision. The day is still traded and flagged.

3.4 Mediator

Agent weights are $\omega_i = (1 - e_i)(1 - r_i)\, e^{-\gamma \Delta t_i}$ with redundancy r_i (0.05–0.1), data age Δt_i in seconds and $\gamma = 10^{-5}$. The aggregate $S_{\text{final}} = \sum_i d_i s_i \omega_i$ yields BUY if $S_{\text{final}} > 0.05$, SELL if $S_{\text{final}} < -0.05$ and NEUTRAL otherwise. Signals map to actions $+0.5$, -0.5 and 0 (buy with half the cash, sell half the coins, hold), the sizing CryptoTrade uses for its rule baselines.

4 Experimental Setup

Data and periods. We use CryptoTrade's public BTC dataset: daily prices (CoinMarketCap), six on-chain statistics (Dune) and up to five filtered news articles per day (Google News). We keep the paper's three test periods: bull (2023-10-01 to 12-01, 61 trading days, BTC +39.7%), sideways (06-17 to 08-25, 69 days, -0.8%) and bear (04-12 to 06-16, 65 days, -15.6%) as the *test* set. Three ranges of the same dataset that the paper does not use form a *development* set: Dev1 (2023-02-20 to 04-11, 50 days, Buy & Hold +21.6%), Dev2 (08-26 to 09-30, 35 days, +3.1%) and Dev3 (12-02 to 2024-01-31, 60 days, +10.8%). Every parameter of the improvements in Section 5.4 is chosen on the development set and the chosen configuration is evaluated once on test. The development set has no sustained bear market.

Baseline reproduction. We reimplement CryptoTrade's environment, four prompts, agent loop and rule baselines. The public code differs from the paper (it disables two of the three technical signals and reflects over three days), so we follow

Method	Bull	Side	Bear	Avg
Buy & Hold	39.66	−0.83	−15.61	7.74
SLMA (rule)	38.53	−3.14	−7.68	9.24
CryptoTrade full	30.47	−1.63	−11.72	**5.71**
CryptoTrade market-only	−3.31	1.76	−8.49	−3.35
Ours full	16.14	1.62	−16.63	0.38
w/o debate	13.53	1.60	−18.36	−1.08
Market + Sentiment	12.00	2.57	−12.55	0.67
Market + Financial	2.12	**6.51**	−18.62	−3.33
Market only	3.09	2.61	−10.01	−1.44

Table 1: Total return (%) on the three BTC periods with the same local LLM.

the paper: three signals and seven-day reflection. Running the original code side by side, our six rule strategies match its returns and Sharpe ratios exactly on all three periods, and therefore also match the paper's Table 2; the four prompts are identical over 12 steps for the same action sequence. On our local model, CryptoTrade reaches +30.47% / −1.63% / −11.72% versus +26.35% / −4.07% / −11.72% reported with GPT-4, so the reproduction is faithful and the model is a capable backbone.

Environment and model. Both systems trade in CryptoTrade's environment: \$1M split evenly between cash and BTC, one action in $[-1, 1]$ per day, no shorting, 0.4% fee plus fixed gas, execution at the open. Both use `gemma-4-26b-a4b` (4-bit QAT) served locally by LM Studio on a MacBook M1 Pro, temperature 0, with all calls cached for exact reruns. At the open of D both systems see prices up to that open and on-chain data and news of $D-1$; a unit test confirms the inputs coincide with CryptoTrade's on all 61 bull days.

Configurations and metrics. We compare CryptoTrade *full* (all sources and reflection), CryptoTrade *market-only* (technical signals only), our full system, and ablations derived offline from the same run by re-aggregating saved opinions: without debate, Market only, Market+Financial and Market+Sentiment. We report total return, Sharpe ratio (Sharpe, 1966) (mean over standard deviation of daily returns, as in CryptoTrade) and directional accuracy, the share of trading days whose direction matches the open-to-open move.

Added to Market only	Bull	Side	Bear
+ Sentiment (news)	+8.9	−0.0	−2.5
+ Financial (on-chain)	−1.0	+3.9	−8.6
+ both, with debate	+13.1	−1.0	−6.6

Table 2: Return change (percentage points) from adding sources to the Market-only configuration.

5 Results

5.1 RQ1: explicit versus implicit conflict handling

Table 1 shows that our full system averages +0.38%, above CryptoTrade market-only (−3.35%) but below CryptoTrade full (+5.71%) and Buy & Hold (+7.74%). It wins only in the sideways period. The ranking repeats on the development set: CryptoTrade full +11.95% (Dev1/2/3: +32.33/−4.62/+8.13), Buy & Hold +11.85%, CryptoTrade market-only +8.83%, ours +8.60% (+22.31/−2.53/+6.01). In the bull period the two systems have almost the same directional accuracy (55.3% vs. 55.1%) yet our return is about half: CryptoTrade chooses its own position size and typically commits 0.8–1.0 when confident, while our system trades a fixed half, and with no shorting every SELL in a rally forfeits gains. Sizing is not the whole story, however: signal-based sizing does not help (Section 5.4), and on Dev1 CryptoTrade is also more accurate (57.1% vs. 50.0%). Explicit conflict handling, in its current form, does not beat implicit fusion, and this holds on two disjoint sets of dates.

5.2 RQ2: source value and agent signal quality

Table 2 isolates each source. News helps strongly in the bull period and slightly hurts in the bear period. On-chain data helps in the sideways period, where Market+Financial reaches the best directional accuracy of all configurations (63.0%), but costs 8.6 points in the bear period. In the bear period, Market only is our best configuration: every added source makes losses worse.

These ablation differences, however, do not survive a direct per-day measurement on the development set (Table 3). All three agents are near chance (45–48%; with 84–116 directional calls per agent the 95% interval is about ±10 points), and no regime pattern repeats: Market is right on 68.8% of dev down days but 43.8% of test down days. What does repeat is a directional bias. The Senti-

	Accuracy, dev			Acc.,	BUY share
Agent	Up	Down	All	test	dev / test
Market	44.1	68.8	47.6	52.4	42 / 39
Financial	52.3	52.6	48.3	53.9	50 / 49
Sentiment	48.8	41.7	45.4	45.3	**79 / 81**

Table 3: Per-agent directional accuracy (%) by trend regime and share of BUY among directional calls. Regime: up if the open is above MA50 and the 20-day return exceeds 5%, down if below MA50 and under −5%, flat otherwise.

	Bull	Side	Bear
Debate days	36/61	32/69	31/65
Conflict before → after	.63→.24	.61→.22	.60→.22
Flagged (rejected update)	10	6	8
Return gain vs. no debate	+2.6	+0.0	+1.7
Final BUY / SELL / NEUTRAL	19/28/14	27/23/19	26/24/15

Table 4: Debate statistics and final signal distribution of our full system.

ment agent calls BUY about 80% of the time while the price rises on about half of the days, because crypto news is upbeat by default. The Market agent leans to SELL: it follows CryptoTrade's indicator labels literally, and two of them are inverted with respect to the usual convention (kept from the original code): a 15-day average above the 20-day one, and MACD above its signal line, are both labelled "sell". On the 180 days the open is above both EMA20 and EMA50, both labels read "sell" on 47% of days, and the Market agent says SELL on 49% and BUY on 21% of them. Stronger signals are not more accurate either (41–46% in every $|S_{\text{final}}|$ bucket on dev), and longer horizons do not help: against a chance level with the same BUY share, every agent's, our final decision's and CryptoTrade's edge at 1, 3 and 7 days lies between -2.3 and $+1.3$ points pooled.

5.3 RQ3: debate reduces conflict but rarely resolves it

Debate fires on 99 of 195 days (Table 4) and cuts the mean conflict score from about 0.61 to 0.22. It never hurts returns. Yet it changes the final signal on only 11 days, always from BUY or SELL to NEUTRAL, and on none of them does a wrong call become right or vice versa: the gain comes from staying out on uncertain days. The cause is by design: debate decays the strength of contested opinions but never flips their direction, so opposing beliefs shrink towards zero together. The conflict score falls because the agents become less certain, not because they agree. Debate thus acts as a risk filter rather than an error-correction mechanism, a distinction that a lower conflict score alone would hide. Lifting the restriction does not change this (Section 5.4): allowed to flip, agents almost always restate their position.

Does resolving a conflict make the call more accurate? On the 170 days with at least one BUY and one SELL, one side is always right, yet the current system is right 48.3% of the time, the Mediator without debate 47.5% and a majority vote 46.2%. Following Financial gives 55.4% and following Sentiment 41.0%, on both splits (this table was computed after the test periods had been seen). A blind audit of 30 dev debate days (90 opinions checked against their snapshots and full articles) finds only 5 minor factual slips and none that drives a signal: most conflicts are different readings of correct facts, so a debate that adjudicates by checking facts has little to correct.

Variant	Dev	Test	Bull	Side	Bear
Ours (eval-v1)	8.60	0.37	16.14	1.62	−16.63
Trend filter	8.49	**9.04**	36.60	5.13	−14.61
Regime weights	**15.97**	0.00	12.69	1.51	−14.19
Weights + filter	10.84	2.68	18.81	4.19	−14.96
Signal sizing	10.91	−0.55	12.36	1.99	−15.99
All three	13.15	3.62	21.18	9.23	−19.55
Bias correction (offline)	8.38	1.28	15.57	3.89	−15.63
Longer trend context	8.67	–	–	–	–
Debiasing prompts	2.47	–	–	–	–
Debate may flip	8.56	–	–	–	–
Never SELL	11.22	7.39	38.55	−1.62	−14.75
Reliability arbiter	9.00	−0.15	12.00	4.63	−17.07
CryptoTrade full	11.95	5.71	30.47	−1.63	−11.72

Table 5: Average return (%) of improvements: each is selected on the development set (Dev) and evaluated once on test. The three variants that need new LLM runs were not run on test because they did not improve on dev.

5.4 RQ4: improvements tuned on development data

We test six improvements (Table 5). Three change only aggregation and are recomputed from saved opinions: a *trend filter* that blocks SELL in an up-trend and BUY in a down-trend (18 settings: MA20/50, 10/20/30-day return, 0/3/5% band); *regime weights* that multiply the Financial and Sentiment contributions by $\{0, 1, 2\}$ per regime (729 combinations per trend setting); and *signal sizing* $\text{clip}(S_{\text{final}}/k, -1, 1)$, $k \in \{0.1, 0.2, 0.3, 0.5\}$. *Bias correction*, also computed offline, replaces each agent's score $d_i s_i$ by its deviation from the agent's

own mean over the previous $K \in \{10, 20, 30\}$ days, i.e. whether today's view is better or worse than usual. *Longer trend context* changes the Market agent's input and needs a new LLM run: the agent also sees the 30-day return, the distance to MA50 and the number of consecutive opens on that side of MA50.

Following the protocol, the dev winner is regime weights (+15.97%), which falls to +0.00% on test. Leaving one dev range out and selecting on the other two confirms the overfitting risk of a grid of thousands of combinations on 145 days. The trend filter is the only variant above CryptoTrade on test (+9.04%), but it was not selected: every one of its 18 settings is *below* eval-v1 on dev (+1.1 to +8.5%) and *above* it on test (+2.6 to +10.6%). Its gain is a property of the test bull period, a long steady rally in which blocking SELLs simply holds BTC (+36.6% vs. Buy & Hold +39.7%), not of its parameters. Longer trend context makes the Market agent take more positions (112 vs. 84) without making it more accurate (49.1% vs. 47.6%); it still calls BUY only 41% of the time. Bias correction balances the BUY share (Sentiment 79%→53%, Market 42%→51%) but leaves accuracy unchanged (final 45.5% vs. 45.0%). *Debiasing prompts* attack the bias at its source: the Sentiment agent also reads the headlines of the three earlier days and is told that routine optimism is not a buy signal, and the Market agent is told to take a side only when most indicators agree. Sentiment's BUY share falls (79%→65%) but so does its accuracy (45.4%→39.5%); the Market agent abstains on 73% of days (was 41%), and the dev return drops to +2.47%. *Debate may flip* lets each debating agent state a stance, which replaces its direction when the rebuttal strength is at least 0.65 (fixed beforehand from the eval-v1 dev distribution: about 0.58 when the others agree or abstain, 0.80 when both oppose) and its new explanation passes the source check; specialist opinions and RCA are replayed from eval-v1. Over 426 agent-rounds on 71 debate days, agents flip only 5 times, all the Financial agent moving from BUY to NEUTRAL when alone against two SELLs. This is conformity, not correction, i.e. the sycophantic convergence of Huang et al. (2026) on real data: on 3 of those 5 days the price rose, and no flip changes the final decision. Dev return is +8.56% (eval-v1 +8.60%), final accuracy 45.0% in both.

Never SELL comes from a grid of asymmetric rules (BUY at $S_{\text{final}} \geq 0.05$, SELL only at $S_{\text{final}} \leq -k$ or never), motivated by the environment: without short selling, a wrong SELL in a rally costs the gain on the coins sold. Dev selects never selling; on test it reaches +7.39%, above CryptoTrade (+5.71%) but below Buy & Hold (+7.74%), and pooled it is level with CryptoTrade (+0.48 points, $p = 0.77$). The whole gap between the two systems therefore lies in our SELLs, but closing it is a matter of holding more BTC, which is also how CryptoTrade wins (it buys on 91% of its directional days). The *reliability arbiter* settles conflict days by each agent's record on past conflict days. Selected on dev with records reset per window it does not help. With records carried across the year (exploratory), accuracy on all trading days rises on both splits, from 49.6% to 52.3% pooled ($p \approx 0.4$), while returns fall (dev +5.33%, test −1.40%): it mostly rules against the BUY-leaning Sentiment agent and so sells more in a rising year. Combined with the SELL rule it loses its accuracy gain. On BTC accuracy and return trade off; the pre-registered ETH check below refutes the accuracy gain itself.

Are three agents needed? Zeroing the Financial and Sentiment contributions gives a Market-only variant that is *better* on dev (+12.32% vs. +8.60%) and worse on test (−1.59% vs. +0.37%). Pooled over all six ranges, three agents with the filter reach +8.76%, on par with CryptoTrade full (+8.83%) and below Buy & Hold (+9.80%); without the test bull period they fall to +3.20%, below Market only (+5.83%). The value of the extra agents is therefore not established: an always-bullish news agent looks useful only in a long rally.

5.5 Pre-registered check on ETH

Because the arbiter was found after the test periods had been seen and no unused BTC dates remain, we fixed a plan before downloading any ETH data (data, the same six windows, systems, hypothesis and tests; tag `eth-prereg`). CryptoTrade's ETH files, read as its `get_paths('eth')`, reproduce all 15 ETH rule results of its Table 2; both systems run unchanged except for the asset name, with the original indicator labels.

The primary hypothesis fails: the arbiter *lowers* pooled direction accuracy from 52.3% to 46.9% (difference -5.4 points, 95% block-bootstrap CI $[-9.5, -1.4]$). On ETH no agent is more reliable in conflicts (51.3%, 50.0%, 49.5%), the existing debate already resolves conflicts well (61.7% on

ETH, mean return (%)	Dev	Test	Pooled
Ours (eval-v1)	3.73	−0.81	1.46
+ reliability arbiter	−0.26	−3.85	−2.06
+ never SELL	8.25	**3.22**	**5.73**
CryptoTrade full	7.07	−8.67	−0.80
Buy & Hold	**8.77**	2.25	5.51

Table 6: Pre-registered ETH check; the arbiter and the SELL rule keep their BTC configuration.

conflict days), the 27 decisions the arbiter reverses were right 67% of the time, and on the 52 days debate had left the system out of the market the arbiter's forced calls are right 38.5% of the time. The BTC pattern was a post-hoc artefact. Everything else replicates: no agent and not CryptoTrade beats chance at 1, 3 or 7 days; CryptoTrade buys on 83% of its directional days; on uptrend days both inverted labels read "sell" 56% of the time and the Market agent sells on 56%; our action order beats its shuffles ($p = 0.049$). With falling prices in two of three test periods, CryptoTrade's buying habit now costs it (Table 6): our system beats it by 7.85 points on test ($p = 0.072$), and never selling beats it by 11.89 points ($p = 0.002$), on par with Buy & Hold.

5.6 Analysis: hidden aggregation bias

The time-decay penalty treats day-old data as stale: on-chain data and news are from $D-1$, so their factor is $e^{-0.864} \approx 0.42$ against 1 for prices. With equal noise, the Market agent therefore carries about 2.5 times the weight of each other agent, and Sentiment's weight is fixed near $0.5 \times 0.9 \times 0.42 \approx 0.19$. A design choice intended for live freshness becomes a structural bias in a daily backtest. Signals also do not follow the trend: during a 40% rally the system issued 28 SELLs against 19 BUYs, and during the decline 26 BUYs against 24 SELLs, in line with the biases of Table 3.

Statistical tests. A paired moving-block bootstrap (blocks of 5 days, 10,000 resamples) of the mean return difference finds no significant difference at the 5% level: ours minus CryptoTrade full is -5.33 points on test (95% CI $[-18.1, 3.1]$, $p = 0.24$) and -4.34 pooled over six ranges ($p = 0.24$); the trend filter's test gain of $+3.33$ has CI $[-2.5, 9.7]$; and no LLM-rerun improvement differs from eval-v1 on dev (debiasing prompts -6.13, $p = 0.24$). Exact binomial tests find no accuracy different from 50%, including CryptoTrade's own final decisions (49.4%, $n = 253$, $p = 0.90$): its higher return comes from exposure, not from better direction calls. Detecting 55% against 50% with 80% power would need about 780 directional days, three times our data.

Timing. Shuffling each strategy's own actions within each window (5,000 times) separates timing from exposure. Our shuffled actions earn +0.02% on average against +4.49% in the original order, which beats 98.7% of shuffles ($p = 0.013$); CryptoTrade's shuffles already earn +6.32% from buying often, and its order adds the rest ($p = 0.052$). Although neither system predicts direction at 1–7 days, both action sequences follow multi-week trends better than chance, so daily directional accuracy alone is a poor measure of a trading agent.

Cost. Per period, CryptoTrade full needs about 50 minutes and our system about 4.6 hours on the laptop, because of the RCA, debate and verification calls; the monetary cost of both is zero with a local model.

6 Conclusion

Under identical data, environment and model, separating sources into specialised agents and handling their conflict explicitly does not outperform CryptoTrade's implicit fusion, on the paper's test periods or on 145 held-out days. The study yields four lessons for conflict-aware agent design. A debate that only decays confidence lowers measured disagreement without resolving it, and allowing direction changes does not help: the agents almost never change their mind, and when they do, the minority joins the majority. Individual agents can be near chance and systematically biased, the news agent mostly by reading upbeat coverage as a buy signal. When agent signals carry little information, aggregation changes such as regime weights, trend filters and sizing only change market exposure, so their apparent gains depend on the period evaluated; a dev-then-test protocol exposes this where a single test set would not. Finally, aggregation penalties can encode unintended biases, and benchmark label conventions can steer an LLM agent against the trend. Telling the agents about their bias shifts the mix of directions but not their accuracy. Finally, CryptoTrade's edge is exposure, not prediction: its lead on BTC becomes a loss on ETH, where prices fell. Evaluations should therefore separate exposure from timing and prediction, and patterns found after seeing the data need fresh data: our

one post-hoc improvement failed a pre-registered check. Future work should present indicators in a standard convention to both systems and try larger models that may weigh opposing evidence rather than restate their view.

Limitations

Each period has only 35–69 trading days; none of our return or accuracy differences is statistically significant, so "trails CryptoTrade" means "does not beat it", not "is significantly worse". We study two assets of the same year and one 26B-parameter model; results may differ for SOL, other years or larger models. The model may have seen 2023 market data during pretraining; this affects both systems equally but may inflate absolute returns. The test set was used once per selected improvement; we report every such result, but some selection risk remains, and the conflict analysis and long-run arbiter were explored after the BTC test periods had been seen and were then checked on ETH. The development set lacks a sustained bear market. The ± 0.5 action mapping and the parameters of the conflict score, debate and Mediator were set by design intuition. Ablations and aggregation variants are re-aggregations of one run rather than independent runs.

Ethics Statement

This work studies trading signals in simulation and does not constitute financial advice. The CryptoTrade dataset is used under its CC BY-NC-SA 4.0 licence for non-commercial research.

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A Parameters

Conflict Analyzer: $\alpha = 0.6$, $\tau = 0.4$, no-conflict ceiling 0.15. Debate: 2 rounds, $\beta = 0.35$, equal rebuttal weights. Mediator: $\gamma = 10^{-5}\,\mathrm{s}^{-1}$, signal band 0.05, redundancy 0.05 (Market) and 0.1 (Financial, Sentiment). Action mapping BUY/SELL/NEUTRAL $= +0.5/-0.5/0$.

B Directional accuracy and trading days

Method	Bull	Side	Bear
CryptoTrade full	55.1 / 49	45.6 / 68	50.0 / 56
CryptoTrade mkt-only	48.1 / 52	45.2 / 62	56.9 / 58
Ours full	55.3 / 47	56.0 / 50	48.0 / 50
w/o debate	55.8 / 52	54.7 / 53	45.3 / 53
Market + Sent.	52.9 / 51	51.1 / 47	49.0 / 51
Market + Fin.	58.3 / 48	63.0 / 54	45.8 / 48
Market only	54.5 / 44	54.8 / 42	50.0 / 40

Table 7: Directional accuracy (%) / number of trading (non-NEUTRAL) days.